Table 1: SAMap cell-type matches between the rebuilt Guilliams et al. 2022 mouse liver atlas and HLiCA v1.0 (mapping score $\geq$ 0.4; 40 matches). Top gene pairs: GenePairFinder (align_thr 0.1), human/mouse symbol; one name when both agree.

Mouse type	HLiCA type	SAMap score	Top gene pairs (human/mouse)
Hepatocyte			
Hepatocyte	Ribosomal+ Hepatocyte	0.62	*FABP1, APOC3, APOA2, RBP4, ORM1, ORM2/Orm1*
Hepatocyte	Mito+ Hepatocyte	0.55	*SERPINF2/Serpina3k, MST1/F2, MST1/Plg, FGB, FGA, FGL1/Fgb*
Hepatocyte	Periportal Hepatocyte	0.46	*CYP3A5/Cyp3a25, SLC22A25/Slc22a30, CPS1, A1CF, MLXIPL, SLC7A2*
Hepatocyte	Pericentral Hepatocyte	0.46	*SLCO1B3-SLCO1B7/Slco1b2, CYP3A4/Cyp3a25, SLCO1B3-SLCO1B7/Slco1a1, CYP3A43/Cyp3a25, SLC22A25/Slc22a30, IL1RAPL2/Gm20319*
Cholangiocyte			
Cholangiocyte	LAMC2+ Cholangiocyte	0.88	*PKHD1, CTNND2, BICC1, GLIS3, SHANK2, CDH1/Cdh6*
Cholangiocyte	ApoLipo	0.79	*PKHD1, CTNND2, BICC1, GLIS3, THSD4, SHANK2*
Cholangiocyte	Keratin	0.77	*CLDN10/Cldn3, KRT7/Krt8, KRT7/Krt18, CLDN10/Cldn7, TM4SF4, SPP1*
Cholangiocyte	CXCL8+ Keratin	0.77	*KRT7/Krt8, KRT7/Krt18, KRT18/Krt8, KRT8, KRT18, TACSTD2/Epcam*
Cholangiocyte	Mucus Secreting	0.60	*TSPAN8, KRT19/Krt18, KRT19, CLDN4/Cldn7, CLDN4/Cldn3, CLDN3/Cldn7*
Endothelial			
Endothelial	Periportal LSEC	0.80	*STAB2/Stab1, PTPRB, STAB2, OIT3, FLT1/Kdr, FLT1/Flt4*
Endothelial	Portal Vein	0.73	*MECOM, CLEC14A, EFNB2/Efnb1, PTPRB, TM4SF1, EFNB2*
Endothelial	Hepatic Artery	0.66	*PODXL, NHERF2/Slc9a3r2, TIMP3, SOX18, EFNB2/Efnb1, CLEC14A*
Endothelial	Vascular Endothelial Cell	0.63	*PTPRB, TLL1/Nrp1, ENG, TEK, FLT1/Kdr, HSPG2*
Endothelial	Central Venous LSEC	0.58	*OIT3, STAB1, STAB1/Stab2, DNASE1L3, KDR, STAB2/Stab1*
Endothelial	LSEC	0.49	*OIT3, FAM167B, DNASE1L3, KDR, KDR/Flt4, GNG11*
Mesenchymal			
Fibroblast	Portal Fibroblast	0.83	*DCN, COL14A1, LUM/Dcn, PRKG1, DCN/Lrrc4c, LAMA2*
Fibroblast	Hepatic Stellate Cell	0.66	*HGF, CCBE1, PTH1R/Vipr1, HGF/Masp1, COLEC10, PTH1R*
Myeloid			
KC	Kupffer Cells	0.76	*CD5L, CD163/Cd5l, CD5L/Cd163, CD163, SDC3, MSR1/Cd5l*
Monocyte	Non-classical Monocytes	0.71	*MS4A7/Ms4a6c, IFITM2/Ifitm6, S100Z/S100a4, S100A4, MS4A7/Ms4a6b, CTSS*
Monocyte	BAG3+ Monocytes	0.56	*LYZ/Lyz2, S100A9/S100a4, VCAN/Clec4a1, CYBB, S100A4, CTSS*
Monocyte	Activated Monocytes	0.53	*LYZ/Lyz2, SOD2/Ccl9, S100A9/S100a4, CYBB, VCAN/Clec4a1, CCL3L3/Ccl9*
Monocyte	Classical Monocytes	0.42	*S100A9/S100a4, LYZ/Lyz2, VCAN/Clec4a1, S100A12/S100a4, S100A9/S100a6, S100A4*
Neutrophil	Neutrophils	0.81	*CSF3R, CXCR2, CXCR2/Ccr1, MXD1, CXCR2/C5ar1, S100A8*
DC			
cDC1	Type 1 cDCs	0.86	*CLEC9A, WDFY4, XCR1, RAB7B, PPT1, C1orf54/BC028528*
cDC2	DCmac	0.70	*CLEC5A/Clec4b1, CLEC5A/Cd209a, HLA-DPA1/H2-Eb1, HLA-DMB/H2-Eb1, HLA-DPB1/H2-Eb1, CLEC5A/Clec4a2*
cDC2	Type 2 cDCs	0.63	*HLA-DPB1/H2-Eb1, HLA-DPA1/H2-Eb1, HLA-DQA1/H2-Aa, HLA-DQB1/H2-Eb1, HLA-DRB1/H2-Eb1, HLA-DRA/H2-Eb1*
Mig. cDCs	migDCs	0.82	*CCR7, CCL22, IL4I1, FSCN1, CCL22/Ccl17, SLCO5A1*
Transitional DC	pDC	0.41	*CLEC4C/Cd209d, CLEC4C/Cd209a, IRF7/Irf8, BCL11A, IRF8, RUNX2*
pDCs	pDC	0.77	*GZMB/Klk1, P2RY14/Ccr9, PLD4, IRF7/Irf8, CLEC4C/Cd209d, BCL11A*
B/plasma			
B cell	B Cell	0.84	*MS4A1, CD79A, BANK1, PAX5, CD79B, IGHD*
B cell	IgA B cells	0.74	*IGHG1/Igha, IGHG3/Igha, IGHG4/Igha, IGHG1/Iglc1, IGHG1/Igkc, IGHG2/Igha*
B cell	IgG B cells	0.65	*JCHAIN, IGHA1/Igha, MZB1, IGHA2/Igha, IGLC2/Igkc, IGLC2/Iglc1*
B cell	Plasma B Cell	0.43	*JCHAIN, MZB1, IGHG1/Igha, IGHG3/Igha, IGHA1/Igha, IGKC/Iglc1*
T/NK/ILC			
T cell	Regulatory T Cell	0.69	*CTLA4/Cd28, TRAC, CD3D, TRBC2, CD3G/Cd3d, BCL11B*
T cell	Helper T Cell	0.60	*IL7R, CD3D, TRAC, CD3G/Cd3d, BCL11B, LEF1/Tcf7*
T cell	MAIT T Cell	0.53	*IL7R, TRAC, CD3D, CD3G/Cd3d, CXCR6, BCL11B*
NK cell	Dim NK Cell	0.65	*PRF1, GZMB/Gzma, KLRF1/Klra4, GZMB/Cma1, NKG7, GZMH/Gzma*
ILC1	Bright NK Cell	0.66	*XCL1, XCL2/Xcl1, IL2RB, CD160, KLRF1/Klrb1c, KLRF1/Klrb1b*
ILC1	Exhausted CD8 T Cell	0.52	*GZMK/Gzmb, GZMA/Gzmb, CCL5, GZMK/Gzma, GZMH/Gzmb, GZMM/Gzmb*
ILC1	CD8 T Cell	0.46	*GZMK/Gzmb, GZMA/Gzmb, CCL5, GZMH/Gzmb, GZMK/Gzma, GZMA*